

Testing the Tau-Core Observer Principle with Gravitational Lensing: Theoretical Feasibility, Data Audit, and Missing Evidence-Grade Products

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Abstract

The tau-core observer principle is treated here as an operational working hypothesis, not as a completed covariant field theory. In standard lensing language, the paper studies a restricted perturbation of the image-level Fermat surface: an effective latent operator whose coefficient is shared across image systems rather than absorbed into arbitrary per-lens nuisance freedom. The working hypothesis is that a gravitational measurement is not only a property of the source-side mass distribution: it is also a statement about which projection channels are accessible to the observer. Strong gravitational lensing is a natural place to test this principle because a single background source can be observed through several images, each with its own parity, Fermat potential, path geometry, and measured time delay. In this paper we ask a deliberately limited question: is it theoretically possible to turn the tau-core observer principle into a falsifiable gravitational-lensing test, what data have we used to stress that test, and which data products are still missing? The proposed test observable is a shared amplitude, `epsilon_tau`, multiplying the channel `T2_parity_abs_fermat`: a parity-weighted, absolute-Fermat term that must be common across lenses and must remain distinguishable from ordinary lens-model nuisance freedom.

We make three contributions. First, we define the necessary proof gate: a no-T2 baseline must reproduce a published time-delay distance or null posterior from evidence-grade image/model products before any tau-core perturbation is allowed. Second, we show that the proposed T2 channel is mathematically identifiable in controlled synthetic lensing experiments. In a 28-lens hierarchical run with 224 position-constraint rows, an injected `epsilon_tau=0.05` is recovered as `epsilon_hat=0.0499193089`; blind and cached stress tests retain the active T2 branch under core nuisance perturbations. Third, we close a static-lens control branch using HFF cluster morphology controls. These controls do not prove T2, but they show that the workflow does not automatically manufacture tau-like signals in systems without a time-delay posterior.

The present paper does not claim that the observer principle has been proven in real lensing data. Instead, it establishes the theoretical test pathway and the current data gap. A no-contact public-product audit found no target with

enough public image/model product depth to support the required no-T2 reproduction gate. HE0435-1223 public imaging can be ingested, but the public-only PSF/model route fails validation; additional probes of RXJ1131-1231, WFI2033-4723, WGD2038-4008, and DES J0408-5354 remain product-blocked. The result is therefore a feasibility-and-reproducibility paper: the lensing test is well-defined and stress-tested, but the evidence-grade public products needed for a real-data proof are still missing.

Author and AI-Assistance Statement

This manuscript is an auditable methodological draft. The originating author is an independent researcher and does not claim institutional expertise in gravitational lensing, cosmography, or modified gravity. The hypothesis, test priorities, and research direction originated from the author and were developed with AI assistance into scripts, tables, summaries, and prose. AI assistance is not peer review or expert validation. The work should be evaluated by reproducibility, conservative claim boundaries, source traceability, and independent expert audit.

1. Introduction: The Tau-Core Observer Principle

The tau-core observer principle begins from a simple measurement distinction. In a gravitational system there is a physical source, such as a galaxy, cluster, or lensing mass distribution. But an observer never measures the source in itself. The observer measures light paths, arrival times, image positions, frequency shifts, projected shapes, and other accessible channels. The tau-core idea is that some gravitational residuals may live not in a hidden mass component alone, and not in a source-only modification alone, but in the mapping between source geometry and observer-accessible projections.

In this language, **tau** is not introduced as an extra directly observed clock that can simply be read off from data. In the present paper, **tau** is treated as an operational latent projection coordinate: no claim is made here that it is a covariant field, a gauge-invariant degree of freedom, or a completed dynamical sector. The operational question is therefore not “can we see tau directly?” but:

Can we find an observer-accessible channel whose geometry requires a shared tau-like projection amplitude, after ordinary source/lens nuisance freedom has been controlled?

This is why gravitational lensing is attractive. A lens does not give one measurement of one source. It gives multiple images of the same source through different light paths. In a time-delay lens these paths have different arrival times and different Fermat-potential structure. In a quad lens they also have a rich image-parity structure. Thus, a lens is almost a natural interferometer for observer-projection ideas: several observed channels constrain one underlying source-lens configuration.

The tau-core observer principle would become physically meaningful in lensing only if it predicts a constrained cross-image or cross-lens structure. If every lens were allowed to choose its own arbitrary correction, the idea would be empty. The test must therefore look for a shared amplitude, common across lenses or image systems, attached to a geometrically defined channel.

In this paper the candidate channel is:

`T2_parity_abs_fermat`

It combines three ingredients that are natural in time-delay lensing:

1. image parity;
2. Fermat-potential differences;
3. relative time-delay structure.

The associated amplitude is:

`epsilon_tau`

The goal of the present paper is not to claim that `epsilon_tau` has been measured. The goal is narrower and, at this stage, more defensible:

1. formulate the lensing test implied by the observer principle;
2. show that the test is identifiable in synthetic lensing systems;
3. show that static controls do not automatically generate false positives;
4. audit which public data products were actually tested;
5. state precisely which evidence-grade products are still missing.

With that framing, the paper is a theoretical feasibility and data-readiness study for a future proof, not a proof claim itself.

2. Toward a Physical Theory: Minimal Effective Formulation

The largest weakness of the present theory is also the reason this paper is framed as a proof-gate manuscript rather than as a modified-gravity discovery paper. The tau-core observer principle is not yet supplied with a complete physical definition. However, the idea can be made more precise than a philosophical statement. This section separates three levels. Level 0 is the measurement principle; Level 1 is the phenomenological tau-lensing model used in this paper; Level 2 is a complete covariant tau-core gravitational theory.

The present paper works at Level 1. It uses Level 0 as motivation and treats Level 2 as an open theory-completion problem.

2.1 Measurement Principle

The measurement principle is that an observer does not access the gravitational source directly. The observer accesses a finite set of projection channels gen-

Tau-core observer principle in a multiply imaged time-delay lens

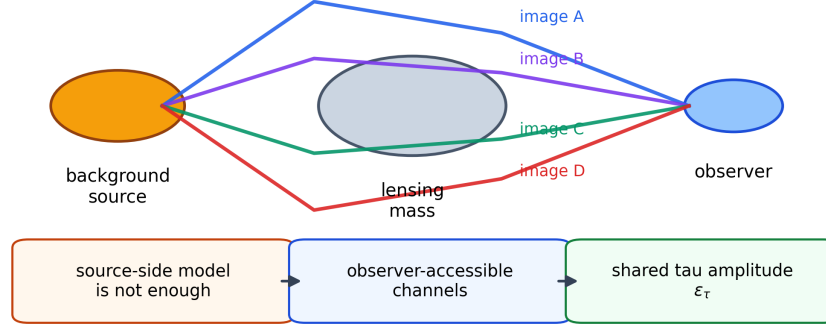


Figure 1: Schematic of the tau-core observer principle in a multiply imaged time-delay lens. The same background source reaches the observer through several image channels, making parity, Fermat structure, and time delays observer-accessible probes rather than source-only quantities.

erated by null geodesics, arrival-time surfaces, image parities, redshifts, and detector conventions.

The tau-core claim is that some residual structure may be attached to this observer-accessible projection map rather than to a new source-side mass component alone. In other words, a successful model must not merely change the mass distribution; it must predict a constrained correction to the map between the source-lens geometry and the observed channels.

This principle is still broad. It becomes physical only after we specify what object carries the correction, how it transforms, how it couples to light or to the lensing potential, and how the GR limit is recovered.

2.2 Relational Observable Ansatz

The minimal phenomenological ansatz used here is not a full field theory, but it identifies the object that a future field theory would have to reproduce. We introduce a latent relational projection variable:

$$\tau(x; u).$$

The notation is deliberately conservative. The paper does not decide whether τ is a scalar field, foliation variable, hidden clock, boundary degree of freedom, information variable, or gauge artifact. It treats $\tau(x; u)$ as a relational

observable ansatz: $\mathbf{x}$ labels the spacetime point or lens-plane location used by the model, while $\mathbf{u}$ labels the convention by which an observer-accessible projection is constructed.

The symbol $\mathbf{u}$ should not be read as a new preferred cosmic frame. It is a bookkeeping label for a gauge-fixed observer construction, analogous in spirit to choosing a tetrad, screen basis, source-plane convention, reference image, or detector frame before reporting lensing observables. A completed covariant theory would have to show how this construction transforms when the observer description is changed and which combinations are invariant. The present paper uses $\mathbf{u}$ only to mark that the proposed residual is relational rather than source-only.

This avoids the strongest premature interpretation. We are not postulating a Lorentz-violating aether field, a universal foliation, or a directly observable hidden clock. Those possibilities are not ruled out by the notation, but they are not claimed here. The only object used in the lensing test is the first-order projection of this relational structure onto image-level Fermat and parity data.

At the effective lensing level, the paper does not require the full function $\mathbf{tau}(\mathbf{x}; \mathbf{u})$. It requires only the first-order image-channel projection:

$$\delta\Phi_i = \epsilon_\tau T_i^{(2)}, \quad T_i^{(2)} = \pi_i |\Delta\phi_i|.$$

Here Φ_i is the effective arrival-time/Fermat structure for image i , $\Delta\phi_i$ is the model-derived Fermat-potential difference relative to a reference image, and π_i is the image parity or parity proxy. This is the bridge between the broad observer principle and the actual data test.

2.3 Tau-Core Projection Kernel

The older tau-core/TPG construction treated the observable correction as a projection-window integral over a hidden or relational time coordinate, not as a directly visible matter component. In its simplest form, a projected observable correction has the structure

$$\mathcal{P}[X] = \int_0^{T[X]} w(\tau) d\tau,$$

where X is the ordinary observable that sets the depth of the projection window. The tau-core symmetry assumption is dilation covariance on positive τ -space:

$$w(\lambda\tau) d(\lambda\tau) = w(\tau) d\tau.$$

Up to regularisation at the origin, this selects the scale-invariant density $w(\tau) \propto 1/\tau$. The regularised minimal representative is

$$w(\tau) = \frac{\alpha}{1 + \tau}.$$

Therefore a dimensionless projection depth T produces

$$\mathcal{P}(T) = \alpha \int_0^T \frac{d\tau}{1 + \tau} = \alpha \ln(1 + T).$$

For lensing, the natural image-level candidate for a dimensionless projection depth is the Fermat offset normalized by a lens time-delay scale:

$$T_i = \frac{|\Delta\phi_i|}{\phi_\star}.$$

In the small-window limit $T_i \ll 1$,

$$\mathcal{P}(T_i) = \alpha T_i + \mathcal{O}(T_i^2) = \frac{\alpha}{\phi_\star} |\Delta\phi_i| + \mathcal{O}(|\Delta\phi_i|^2).$$

This supplies the tau-core origin of the absolute Fermat factor: it is the leading term of a dilation-covariant projection-window integral when the projection depth is the image's Fermat offset. The remaining sign/orientation information cannot come from T_i , because T_i is a magnitude. The available local orientation invariant of the arrival-time surface is the Morse parity $\pi_i = \text{sgn det } H_i$. Combining the tau projection magnitude with Morse orientation gives

$$\delta\Phi_i = \epsilon_\tau \pi_i |\Delta\phi_i| + \mathcal{O}(\epsilon_\tau |\Delta\phi_i|^2, \epsilon_\tau^2).$$

This is still not a complete derivation from a covariant action. It is, however, a more specific tau-core physical route: dilation covariance supplies the projection kernel, the Fermat offset supplies the lensing projection depth, and Morse parity supplies the local orientation sign.

2.4 Effective Action Template

A complete theory would require an action. This paper does not claim to have one. The minimal action template that would be compatible with the present lensing test has the schematic form:

$$S = S_{\text{GR}}[g] + S_{\text{matter}}[g, \psi] + S_\tau[g, \tau, u] + \epsilon_\tau S_{\text{coupling}}[g, \tau, u, k].$$

where $\mathbf{g}$ is the metric, $\mathbf{psi}$ denotes ordinary matter fields, $\mathbf{u}$ is the gauge-fixed observer/projection construction, and $\mathbf{k}$ is the photon/null direction. The coupling term must reduce, in the weak-field thin-lens limit, to an image-channel correction proportional to `parity_i * |Delta phi_i|`.

This template does not define the dynamics. It defines a consistency target: any future tau-core action that wants to explain the lensing T2 channel must produce the same first-order effective observable, while also satisfying covariance, conservation, and the GR limit.

2.5 Arrival-Time Functional Route

A more concrete variational route can be stated at the lensing-functional level. In ordinary time-delay lensing, the image branches are stationary points of an arrival-time functional. The tau-core correction can be represented phenomenologically as an additional branch functional:

$$\mathcal{A}_{\text{eff}}[\theta] = \mathcal{A}_{\text{GR}}[\theta] + \epsilon_\tau \mathcal{A}_\tau[\theta; u].$$

For the weak-field thin-lens reduction, the GR part is proportional to the Fermat potential,

$$\mathcal{A}_{\text{GR},i} \propto \phi_i.$$

The tau-core ansatz is that the correction does not add a new projected mass component directly. Instead, it weights the observer-accessible branch by the tau projection kernel and by the local Morse orientation of the stationary point:

$$\mathcal{A}_{\tau,i} = \pi_i \phi_\star \mathcal{P}\left(\frac{|\Delta\phi_i|}{\phi_\star}\right).$$

Using the dilation-covariant kernel from the previous subsection gives

$$\mathcal{A}_{\tau,i} = \pi_i \phi_\star \alpha \ln\left(1 + \frac{|\Delta\phi_i|}{\phi_\star}\right).$$

For $|\Delta\phi_i|/\phi_\star \ll 1$,

$$\mathcal{A}_{\tau,i} = \alpha \pi_i |\Delta\phi_i| + \mathcal{O}\left(\pi_i \frac{|\Delta\phi_i|^2}{\phi_\star}\right).$$

Absorbing the dimensionless normalization into ϵ_τ yields the T2 perturbation. This gives a concrete variational interpretation of the operator: T2 is the leading branch contribution of a tau-projection term added to the arrival-time functional.

The stationarity condition becomes

$$\nabla_\theta \left(\phi + \epsilon_\tau \pi \phi_\star \mathcal{P}\left(\frac{|\Delta\phi|}{\phi_\star}\right) \right) = 0.$$

In the present paper we do not solve this modified lens equation. The operational test uses the first-order correction evaluated on the baseline GR/lens-model image branches. This is equivalent to a perturbative treatment:

$$\theta_i = \theta_i^{(0)} + \epsilon_\tau \theta_i^{(1)} + \mathcal{O}(\epsilon_\tau^2),$$

while the current T2 design keeps only the branch-level shift in the arrival-time/Fermat observable. A full theory would need to propagate this correction consistently into image positions, magnifications, time delays, and posterior lens parameters.

2.6 Symmetry, Covariance, and Conservation Requirements

A physically acceptable Level-2 theory should satisfy at least the following requirements:

Requirement	Minimal condition
diffeomorphism covariance	the full action must be coordinate independent
observer-construction covariance	changes of gauge-fixed observer/projection convention must have a defined transformation law
GR limit	all tau corrections vanish or become unobservable as <code>epsilon_tau -> 0</code>
conservation	modified field equations must imply a consistent stress-energy balance
lensing consistency	null propagation must remain well-defined and causal
no arbitrary per-lens freedom	the tau amplitude must be shared or hierarchically constrained

In the present paper, only the last two operational requirements are directly tested. The others are theory-completion gates.

2.7 Gauge Meaning

The gauge problem is central. Fermat potentials, image labels, and parity assignments are not raw detector counts; they are products of a lens model. Therefore T2_i cannot be treated as gauge-invariant merely because it is computable. A defensible empirical claim would require fixed image-label and reference-image conventions, a reproducible lens-model posterior, parity derived from the same model family, Fermat differences propagated through posterior uncertainty, and stability of the T2 result under allowed reparameterizations.

Until this is done, the tau-lensing model is best read as a gauge-controlled phenomenological perturbation, not as a fundamental observable.

2.8 What Is Still Open

The Level-2 theory remains incomplete. In particular, this paper does not yet provide:

Required theory ingredient	Present status
covariant action	template only
field content	relational projection ansatz <code>tau(x; u)</code> only
covariance structure	required but not derived
defining symmetry	required but not derived
conservation relation or Ward identity	required but not derived
derivation of the GR limit	not derived; imposed operationally by <code>epsilon_tau -> 0</code>
gauge meaning of <code>tau</code>	partially specified as a model-convention stability requirement

This is a hard claim boundary. At present, the protocol is intentionally stronger than the theory. This is a feature of the paper’s claim boundary: it turns an under-specified principle into a falsifiable data requirement instead of promoting it as a finished theory.

The minimal object defined in this paper is therefore not a new action. It is an operational perturbation test on top of an ordinary lensing baseline. The baseline lens model supplies image parity, Fermat structure, and time-delay constraints; the observer-channel design vector is `T2_parity_abs_fermat`; the shared scalar amplitude is `epsilon_tau`; and the GR/null limit is recovered when $\epsilon_\tau = 0$.

In this operational version, `T2_parity_abs_fermat` is a predefined design vector computed from lens-model products, and `epsilon_tau` is a shared scalar coefficient. The test asks whether a single shared coefficient can account for structured residuals after a no-T2 baseline and nuisance freedom have already been controlled. This is not yet a derivation from first principles. It is a falsifiable bridge between the observer principle and data.

The paper’s theory-status statement is therefore: a complete tau-core field theory is not yet supplied; the tau-lensing perturbation test is defined only at phenomenological Level 1; and real-data evidence is unavailable until a no-T2 reproduction is achieved.

3. Bridge to Standard Lensing and Inference Language

The terminology used above is intentionally internal to the tau-core program, but the statistical object is familiar. The proposed test can be read in four standard languages.

Theory-completion gap vs operational lensing protocol

field-theory completion gates		operational lensing test gates	
covariant action	open	baseline no-T2 gate	required
field content	open	T2 design vector	defined
covariance structure	open	shared epsilon_tau	defined
symmetry	open	synthetic identifiability	passed
conservation relation	open	static false-positive controls	closed
GR limit	imposed, not derived	real-data proof	blocked
Claim boundary: the paper defines a falsifiable observer-channel test; gauge means it does not yet define a complete tau-core gravitational theory.			

Figure 2: Figure 2. Theory-completion gap. The operational observer-gated lensing protocol is defined, while the field-theory ingredients needed for a complete physical tau-core model remain open gates.

First, in lensing perturbation language, the model is a constrained perturbation of the Fermat surface. A conventional time-delay lensing analysis models image positions, magnifications, and delays through a lens potential and a Fermat potential. The tau-lensing term does not replace that baseline. It asks whether a specific first-order residual term, tied to parity and Fermat geometry, is favored after the baseline model has already reproduced the published time-delay-distance or null posterior:

$$\Phi_i = \Phi_i^{\text{GR+lens}} + \epsilon_\tau O_i + N_i, \quad O_i = \pi_i |\Delta\phi_i|.$$

Second, in effective-operator language, `T2_parity_abs_fermat` is an empirical operator on image-level lens-model products. The operator is not arbitrary: it uses quantities already present in time-delay lensing, and it is tested against nearby controls that should not be promoted if the effect is merely absorbing modeling freedom. However, it is not yet theoretically inevitable. A complete dynamical theory would need to explain why parity, absolute Fermat structure, and a shared scalar coefficient are selected rather than another operator in the same image-level family.

Third, in Bayesian hierarchical language, `epsilon_tau` is a population-level coefficient. Individual lenses may have their own ordinary nuisance parameters, but the tau coefficient is not allowed to float independently for every lens or image. This is the key statistical distinction between a physical shared effect

and a flexible residual-fitting term.

$$d_l \sim p(d_l \mid \theta_l, \eta_l, \epsilon_\tau), \quad \epsilon_\tau \sim p(\epsilon_\tau), \quad (\theta_l, \eta_l) \sim p_l(\theta_l, \eta_l).$$

Fourth, in latent-variable language, **tau** is not directly observed. It is a latent relational variable whose only allowed appearance in this paper is through a pre-defined image-channel projection. This is why compressed time-delay-distance posteriors are insufficient: the latent projection test requires access to image labels, parities, Fermat differences, and model uncertainties upstream of the final cosmographic posterior.

This bridge also clarifies what the paper does not claim. It does not claim a new mass component, a direct detection of a new field, or a replacement for standard time-delay cosmography. It defines a disciplined perturbation term and asks whether future evidence-grade lensing products can falsify it.

4. Why Lensing Is the Right Test Bed

The tau-core observer principle is an attempt to formalize a simple intuition: an observer does not see a gravitational source directly; the observer sees a projection of source geometry through accessible measurement channels. In ordinary weak-field applications this distinction can be hidden inside an effective potential. Strong lensing makes the distinction sharper. A lensed quasar, for example, is observed through multiple images of one source. These images share the same lens but differ in parity, path geometry, Fermat potential, arrival time, and local image constraints.

This makes time-delay lensing a natural test bed. If the observer principle has a lensing signature, it should not appear as an arbitrary per-lens fudge factor. It should appear as a shared, geometrically structured amplitude that uses image-level information and survives nuisance controls.

The central question of this paper is whether the tau-core observer principle can be turned into a falsifiable time-delay-lensing test, and what data products are required to run it.

Our answer is yes at the protocol level, but not yet at the real-data proof level. We define the test criterion, test its identifiability synthetically, audit public product availability, document the data products actually used, and close a static-control branch.

5. The Observer Principle in a Lensing System

For a time-delay lens, a standard lens model gives image positions, time-delay relations, and Fermat-potential differences. Let image *i* have a parity indicator and a Fermat-potential difference relative to a reference image. The tau-core observer principle predicts that an admissible observer-channel correction should depend on the multi-image geometry, not merely on a scalar distance posterior.

We therefore define the candidate T2 channel:

`T2_parity_abs_fermat`

The name is intentionally operational:

- **parity**: the term is sensitive to image parity or a model-derived parity proxy;
- **abs_fermat**: the term uses absolute Fermat-potential structure rather than delay-only compression;
- **T2**: the term is the only synthetic branch promoted as an active tau-core candidate in this work.

The amplitude multiplying this channel is:

`epsilon_tau`

A crucial constraint is that `epsilon_tau` must be shared. If every lens, image, or exposure receives its own free amplitude, the term is not an observer principle; it is ordinary nuisance absorption.

6. Operator Origin and Model-Comparison Requirement

The current paper does not derive `T2_parity_abs_fermat` from a dynamical principle. It motivates it as the simplest candidate operator that uses three features special to time-delay lensing:

Ingredient	Motivation	Remaining weakness
image parity	distinguishes minima, saddles, and maxima of the arrival-time surface	no symmetry has yet been shown to force parity weighting
absolute Fermat structure	keeps sensitivity to image-level arrival-time geometry rather than only compressed delays	no variational derivation yet selects $ \Delta\phi $
shared scalar amplitude	prevents the term from becoming per-lens residual fitting	no completed theory yet predicts a universal scalar coefficient

Thus T2 is a first candidate in an operator family, not a final theoretical necessity. A future dynamical tau-core theory would have to supply at least one of the following:

1. a symmetry argument that selects parity-weighted Fermat structure;
2. a weak-field expansion of a covariant action whose leading lensing correction reduces to T2;
3. a conservation or reciprocity relation that forbids nearby operators;

4. a hierarchical evidence comparison showing that T2 is favored over a pre-defined operator family after nuisance freedom is included.

For the present paper, the conservative statement is:

T2 is a symmetry-motivated candidate operator, not a derived operator.

The necessary future comparison set should include at minimum:

Operator	Role
$O_i^{(0)} = 0$	standard no-tau baseline
$O_i^{(1)} = \Delta\phi_i$	ordinary signed Fermat perturbation
$O_i^{(2)} = \pi_i \Delta\phi_i $	current T2 candidate
$O_i^{(3)} = \Delta\phi_i $	parity-free absolute-Fermat control
$O_i^{(4)} = \pi_i$	parity-only control
$O_i^{(5)} = L_i$	geometric path-length proxy
scrambled $O_i^{(2)}$	label/parity/Fermat falsification control

Only an evidence comparison against this family can answer the reviewer question “why not another operator?”

7. Conditional Weak-Field Derivation Sketch

The paper can go one step beyond naming T2 as a candidate. This section gives a conditional derivation route. It is not yet a theorem of a complete tau-core theory, but it states the geometrical assumptions under which the T2 structure is the leading allowed image-channel correction.

In the thin-lens weak-field limit, image positions are stationary points of the Fermat potential

$$\phi(\theta; \beta) = \frac{1}{2}|\theta - \beta|^2 - \psi(\theta), \quad \nabla_\theta \phi = 0.$$

For image i , define the Hessian

$$H_i = \nabla_\theta \nabla_\theta \phi(\theta_i; \beta), \quad \pi_i = \text{sgn det } H_i.$$

The sign π_i is the Morse parity of the stationary point. Minima and maxima have positive parity, while saddle images have negative parity. Thus parity is not an additional arbitrary label; it is a discrete invariant of the local arrival-time geometry for a fixed lens model.

Now assume that a first-order observer-channel correction obeys four minimal conditions:

1. **Stationary-image locality:** at leading order, the correction is evaluated on each stationary image branch.
2. **Reference-shift invariance:** adding a constant to the Fermat potential cannot change the observable correction, so only differences $\Delta\phi_i = \phi_i - \phi_{\text{ref}}$ may enter.
3. **Tau-window dilation covariance:** the magnitude part of the correction is generated by a projection-window integral with $w(\tau) = \alpha/(1 + \tau)$ and $T_i = |\Delta\phi_i|/\phi_*$.
4. **Arrival-time variational origin:** the same correction can be represented as a branch term in an effective arrival-time functional.
5. **Path-reversal evenness:** the magnitude of the relational delay mismatch is invariant under source-observer reversal, so the leading scalar dependence is even in $\Delta\phi_i$.
6. **Morse-orientation sensitivity:** the correction changes orientation across saddle versus non-saddle branches, so the only local discrete sign available at leading order is π_i .

Under these assumptions, the lowest-order scalar image-channel correction has the form

$$\delta\Phi_i = \lambda \pi_i F(|\Delta\phi_i|) + \mathcal{O}(\lambda^2).$$

If the expansion is analytic in the small dimensionless Fermat offset after normalization by the lens time-delay scale, then

$$F(|\Delta\phi_i|) = c_1 |\Delta\phi_i| + c_2 |\Delta\phi_i|^2 + \dots$$

Keeping only the leading term gives

$$\delta\Phi_i = \epsilon_\tau \pi_i |\Delta\phi_i| + \mathcal{O}(\epsilon_\tau |\Delta\phi_i|^2, \epsilon_\tau^2),$$

which is precisely the T2 operator used in this paper.

This is the first concrete physics bridge in the manuscript. It says that T2 is not selected by numerology alone: it is the leading term if the correction is local on stationary images, invariant under Fermat zero-point shifts, dilation-covariant in the tau-core projection window, representable as an arrival-time functional correction, even under path reversal in magnitude, and sensitive to Morse orientation. The open problem is to derive these assumptions from a covariant action, reciprocity principle, or observer-construction rule rather than imposing them.

8. Minimal Theorem Attempt

This section states the strongest theorem-like claim currently available. It is not a theorem of nature; it is a conditional uniqueness result inside the phenomenological lensing construction.

Definitions. Let ϕ_i be the Fermat potential evaluated on image branch i , let $\Delta\phi_i = \phi_i - \phi_{\text{ref}}$, and let

$$\pi_i = \text{sgn det } H_i$$

be the Morse parity of the image branch. Let an admissible tau-core branch correction be a scalar map

$$\mathcal{Q}_i = \mathcal{Q}(\Delta\phi_i, \pi_i; \epsilon_\tau)$$

added perturbatively to the arrival-time/Fermat observable.

Axioms.

1. **Branch locality.** To first order, $\mathcal{Q}_i$ depends only on the stationary branch i , not on an independently fitted per-lens amplitude.
2. **Fermat zero-point invariance.** $\mathcal{Q}_i$ is invariant under $\phi_i \mapsto \phi_i + C$, so it can depend on $\Delta\phi_i$ but not on the absolute zero of ϕ .
3. **Tau projection depth.** The magnitude dependence enters through a dimensionless projection window $T_i = |\Delta\phi_i|/\phi_\star$.
4. **Dilation-covariant tau weight.** The projection kernel is the regularized dilation-covariant density $w(\tau) = \alpha/(1 + \tau)$.
5. **Path-reversal evenness.** The magnitude of the correction is invariant under $\Delta\phi_i \mapsto -\Delta\phi_i$.
6. **Morse orientation.** The only allowed first-order branch sign is the local Morse parity π_i .
7. **Analytic small-window limit.** The correction admits an expansion for $T_i \ll 1$.

Lemma 1: magnitude term. Axioms 3, 4, and 7 imply

$$\mathcal{P}(T_i) = \alpha \ln(1 + T_i) = \alpha T_i + \mathcal{O}(T_i^2).$$

Thus the leading magnitude term is proportional to $|\Delta\phi_i|$.

Lemma 2: sign term. Axioms 5 and 6 imply that the leading branch sign is not $\text{sgn}(\Delta\phi_i)$, because the magnitude is even under path reversal. The remaining local discrete orientation supplied by the arrival-time surface is π_i .

Lemma 3: first-order uniqueness. Under the above axioms, any admissible first-order scalar correction has the form

$$\mathcal{Q}_i = C \epsilon_\tau \pi_i |\Delta\phi_i| + \mathcal{O}(\epsilon_\tau |\Delta\phi_i|^2, \epsilon_\tau^2),$$

where C is a convention-dependent normalization absorbed into ϵ_τ .

Conditional theorem. Within this phenomenological tau-lensing construction, the T2 operator is the unique leading first-order branch correction satisfying branch locality, Fermat zero-point invariance, tau-window dilation covariance, path-reversal evenness, Morse orientation, and small-window analyticity.

The proof is conditional because the axioms are not yet derived from a covariant action. The theorem attempt therefore upgrades T2 from a hand-chosen operator to the leading term of a specified tau-core lensing construction, but it does not yet prove that nature must realize that construction.

Proof gaps. A complete proof would still need to derive branch locality, path-reversal evenness, and Morse-orientation coupling from a deeper source-observer reciprocity principle or covariant action. It would also need to show how the modified stationarity condition propagates into image positions, magnifications, and time-delay posteriors without introducing uncontrolled gauge artifacts.

9. Axiom Origin and Covariant Closure

The conditional theorem clarifies the next physics problem: the issue is no longer the algebraic form of T2, but the origin of the axioms. At present the axioms have different strengths.

Axiom	Current origin	Status
Fermat zero-point invariance	standard lensing convention; only time-delay differences are observable	physically well motivated
Morse orientation	standard stationary-point geometry of the Fermat surface	geometrically well motivated
tau-window dilation covariance analytic small-window limit	inherited tau-core projection symmetry ordinary effective-field/perturbative assumption	tau-core motivated, not covariantly derived acceptable but regime-limited
branch locality	perturbative image-branch assumption	plausible, not derived
path-reversal evenness	candidate source-observer reciprocity principle	open

Axiom	Current origin	Status
Morse-orientation coupling	identification of parity as the branch orientation sign	plausible, but needs a reciprocity or variational theorem

This table is the current physics map. Some ingredients are standard lensing geometry; some come from the tau-core projection kernel; some are still imposed because the covariant completion is absent.

The missing covariant closure would need to define at least four objects:

1. **Primitive field or relational object.** A variable replacing the purely phenomenological $\tau(x; u)$, together with its transformation law.
2. **Invariant observable.** A scalar or relational quantity that reduces to the tau projection depth T_i in the thin-lens limit.
3. **Conserved or constrained current.** A relation showing why the tau correction cannot be chosen independently on each image branch.
4. **Reciprocity law.** A source-observer or path-reversal statement explaining why the magnitude is even while the branch orientation is carried by π_i .

A schematic covariant closure would therefore have the form

$$S[g, \psi, \tau, u] \implies \nabla_\mu J_\tau^\mu = 0 \implies \delta\Phi_i = \epsilon_\tau \pi_i |\Delta\phi_i| + \dots$$

in the weak-field thin-lens limit. The present paper supplies only the last arrow conditionally. The first two arrows are the open theoretical task.

This distinction matters because the current theorem attempt could fail in two ways. It could fail empirically if the operator family comparison disfavors T2. Or it could fail theoretically if a covariant completion selects a different reciprocity law, a different projection depth, or a different branch orientation structure.

10. Candidate Reciprocity Principle

The most important missing physical ingredient is a reciprocity principle. The candidate principle proposed here is deliberately narrow: it is a condition on the observer-channel correction in the thin-lens weak-field limit, not yet a full covariant theorem.

Let R denote source-observer reversal or, equivalently at this level, orientation reversal of the image branch in the arrival-time surface. An admissible tau-core branch correction is required to decompose into a scalar projection depth and an orientation factor:

$$\delta\Phi_i = \epsilon_\tau \Omega_i \mathcal{P}(T_i),$$

with the transformation rules

$$R : T_i \mapsto T_i, \quad R : \mathcal{P}(T_i) \mapsto \mathcal{P}(T_i), \quad R : \Omega_i \mapsto -\Omega_i.$$

The first two conditions state why the magnitude is even: the projection depth is a recoverability or accessibility magnitude, not an oriented delay. In the lensing reduction this is

$$T_i = \frac{|\Delta\phi_i|}{\phi_\star}.$$

The third condition states that orientation must be carried by a local orientation pseudoscalar of the image branch. For a two-dimensional Fermat surface, the canonical local candidate is the Morse orientation

$$\Omega_i = \pi_i = \text{sgn det } H_i.$$

Thus the reciprocity candidate gives

$$\delta\Phi_i = \epsilon_\tau \pi_i \mathcal{P} \left(\frac{|\Delta\phi_i|}{\phi_\star} \right).$$

Using the tau-core kernel and expanding for small T_i gives T2 at leading order. This explains, conditionally, why the magnitude is even and why parity carries orientation.

The same principle also suggests why the amplitude should be shared. If ϵ_τ is the coupling of the projection current rather than a per-image nuisance parameter, then different image branches sample the same underlying tau-accessibility flux. The covariant version would require a current or constraint such as

$$\nabla_\mu J_\tau^\mu = 0 \quad \text{or} \quad \mathcal{C}_\tau[g, \tau, u] = 0,$$

whose thin-lens reduction leaves one shared coefficient rather than independent branch amplitudes.

This is still a candidate principle. The missing theorem is:

$$\text{covariant reciprocity} \implies R : T_i \mapsto T_i, \quad R : \Omega_i \mapsto -\Omega_i, \quad \Omega_i = \pi_i.$$

Until that theorem exists, path-reversal evenness, Morse-orientation coupling, and shared amplitude are physically motivated but not derived.

11. Reciprocity Theorem To Be Proven

The central theory task can now be stated more sharply. The desired theorem is not “T2 fits data.” It is a covariant statement explaining why the thin-lens observer-channel correction must decompose into an even projection magnitude and an oriented Morse factor.

Candidate theorem statement. Let $\mathcal{O}_\tau[g, \tau, u; \gamma]$ be the detector-accessible tau-core observable associated with a null path γ , where g is the metric, τ is the relational projection object, and u denotes the observer construction. Suppose there exists a source-observer reversal map R such that:

$$\mathcal{O}_\tau[R\gamma] = \sigma(\gamma) \mathcal{O}_\tau[\gamma],$$

where $\sigma(\gamma)$ is an orientation character of the null branch. The theorem needed by this program is:

$$\sigma(\gamma_i) \longrightarrow \pi_i = \text{sgn det } H_i, \quad |\mathcal{O}_\tau[\gamma_i]| \longrightarrow \mathcal{P} \left(\frac{|\Delta\phi_i|}{\phi_\star} \right)$$

in the weak-field thin-lens limit.

If this theorem were proven, then the leading branch correction would follow:

$$\delta\Phi_i = \epsilon_\tau \pi_i \mathcal{P} \left(\frac{|\Delta\phi_i|}{\phi_\star} \right) = \epsilon_\tau \pi_i |\Delta\phi_i| + \dots$$

The theorem requires a covariant measurable. The paper currently has only candidates:

Candidate measurable	Thin-lens target	Problem
relational delay functional along γ	$ \Delta\phi_i $	needs invariant definition under observer/source exchange
projection-depth scalar $T[\gamma]$	$ \Delta\phi_i /\phi_\star$	normalization and gauge dependence must be controlled
tau current flux through an observer screen	shared ϵ_τ	no current J_τ^μ is derived
branch-orientation character $\sigma(\gamma)$	π_i	must be tied to Morse parity, not inserted by hand

This theorem can fail in precise ways. If no covariant scalar reduces to $|\Delta\phi_i|$, the magnitude part fails. If the reversal character is not Morse parity, the parity

weighting fails. If no conserved or constrained tau current exists, the shared amplitude fails. If all three fail, T2 remains only a phenomenological operator.

Thus the real theoretical milestone is:

$$\boxed{\text{covariant measurable} + \text{reciprocity map} + \text{orientation character} \implies \pi_i |\Delta\phi_i|}$$

The present paper reaches the right-hand side conditionally. It does not yet derive the boxed implication.

12. Theory Companion Summary and Open Gates

The preceding sections contain the theory needed inside this paper: the observer-channel correction is treated as a conditional, reciprocity-motivated weak-field candidate, not as a completed prediction of the Tau Core parent program. The longer derivation program now lives in the Tau Core theory hub, where it can be developed without overloading the empirical feasibility paper.

At paper level, the relevant conditional structure is:

$$\begin{aligned} & \text{screen-gauge invariance} + \text{reciprocity-even arrival magnitude} \\ & + \text{branch locality} + \text{shared observer-channel amplitude} \\ \implies & \delta\Phi_i = \epsilon_\tau \Sigma_i \mathcal{P}(T_i^{\text{cov}}) + O(\epsilon_\tau^2). \end{aligned}$$

In the weak-field thin-lens limit,

$$\Sigma_i \rightarrow \pi_i, \quad T_i^{\text{cov}} \rightarrow \frac{|\Delta\phi_i|}{\phi_\star},$$

so the candidate leading term becomes

$$\delta\Phi_i = \epsilon_\tau \pi_i \mathcal{P}\left(\frac{|\Delta\phi_i|}{\phi_\star}\right) = \epsilon_\tau \pi_i |\Delta\phi_i| + O(\epsilon_\tau |\Delta\phi_i|^2, \epsilon_\tau^2).$$

The detailed theory companion is the Tau Core theory-hub lensing module. It tracks the longer proof route and the open theory gates:

Gate	Paper status	Theory-hub role
full reciprocity theorem	conditional theorem target	prove or refute the orientation/magnitude decomposition

Gate	Paper status	Theory-hub role
covariant measurable	candidate via Jacobi orientation and arrival-time scalar	derive the detector-accessible invariant
tau-accessibility current	shared-amplitude candidate	derive conservation or a constrained observer-channel law
screen interaction	controlled effective insertion	derive from boundary variation or replace with a better coupling
dynamical closure / GR limit	requirements stated	derive stress-energy, conservation, and decoupling from a parent action

This paper therefore uses the theory only to justify a falsifiable operator family and a data protocol. It does not claim that the parent theory has already forced the T2 operator uniquely.

13. Missing Geometric Necessity

The strongest remaining physics gap is not statistical. It is geometric. The paper defines an observable framework, a candidate operator family, a conditional weak-field route, and a future inference procedure, but it does not yet show that the T2 structure is forced by a deeper geometry.

A stronger theory would need at least one geometrical selection mechanism:

Missing mechanism	What it would need to show
symmetry necessity	a transformation principle that singles out parity-weighted Fermat structure
variational derivation	an action whose weak-field lensing limit produces the T2 operator
reciprocity theorem	a source-observer or path-reversal identity that requires the same correction
caustic/parity theorem	a result tying saddle/minimum parity structure to the residual sign or weight
dynamical selection rule	a reason the shared scalar amplitude is universal or hierarchical

At present, none of these has been derived. This means the physics kernel is embryonic: the paper has a disciplined way to test a candidate geometrical

idea, but it does not yet contain the theorem that would make the candidate inevitable.

The most accurate classification of the work is therefore:

new physics claim: not established
 geometrical necessity: not derived
 phenomenological test kernel: defined
 falsification and inference protocol: strong

This distinction is important. A negative future test would falsify the proposed T2 phenomenology, not every possible tau-core theory. A positive future test would motivate a deeper geometrical derivation, but would not by itself complete the theory.

14. Breakthrough Roadmap

The current manuscript is not the breakthrough. It is the scaffold that states what a breakthrough would have to look like. Three next results would change the scientific status of the program.

The primary theoretical game changer would be a weak-field derivation. If the T2 structure emerged as the leading correction in a controlled weak-field expansion,

$$\Phi_i = \Phi_i^{\text{GR}} + \lambda\pi_i|\Delta\phi_i| + \mathcal{O}(\lambda^2),$$

then the program would move from a phenomenological operator test to a candidate geometrical theory. The crucial point is not merely the appearance of $|\Delta\phi_i|$, but the joint emergence of parity weighting, absolute Fermat structure, and a shared amplitude.

The most attractive geometrical route to such a result may be a reciprocity or parity theorem. A theorem of this kind would connect source-observer exchange, path reversal, or caustic crossing structure to the sign/weight carried by the image parity π_i . The target statement would have the schematic form:

$$\mathcal{G}_{\text{lens}} \implies \delta\Phi_i \propto \pi_i|\Delta\phi_i| \quad \text{at leading order.}$$

The first serious empirical milestone would be real-data operator competition. The decisive comparison is not T2 versus nothing in isolation, but T2 against the operator family $\{O^{(0)}, \dots, O^{(5)}\}$ after marginalizing lens-model nuisance freedom:

$$Z_2 > Z_m \quad \text{for relevant controls } m \neq 2,$$

with stability under prior sensitivity and posterior-predictive checks.

Until one of these results exists, the paper should be read as a disciplined phenomenological and falsification framework, not as a completed physics theory.

15. Likelihood and Evidence Framework

Because the paper uses population-level language, the future real-data test must be stated as a likelihood problem. For lens **l** and image-level data vector **d_l**, let **theta_l** denote ordinary lens parameters and **eta_l** denote nuisance/systematic parameters. For an operator **O_m**, the model is:

$$d_l \sim p(d_l \mid \theta_l, \eta_l, \epsilon_m, O^{(m)}), \quad \epsilon_m \sim p(\epsilon_m), \quad (\theta_l, \eta_l) \sim p(\theta_l, \eta_l).$$

The posterior for a candidate operator is:

$$p(\epsilon_m, \{\theta_l\}, \{\eta_l\} \mid \{d_l\}, O^{(m)}) \propto p(\epsilon_m) \prod_l p(\theta_l, \eta_l) \prod_l p(d_l \mid \theta_l, \eta_l, \epsilon_m, O^{(m)}).$$

The evidence for the operator is:

$$Z_m = \int p(\epsilon_m) \prod_l p(\theta_l, \eta_l) \prod_l p(d_l \mid \theta_l, \eta_l, \epsilon_m, O^{(m)}) d\epsilon_m d\theta_l d\eta_l.$$

The relevant comparison is not simply whether **epsilon_m** is nonzero. It is whether the T2 operator improves predictive and evidential performance after the Occam penalty for the shared coefficient and after marginalizing ordinary lens-model freedom. The no-tau comparison is:

$$B_{20} = \frac{Z_2}{Z_0}.$$

The operator-competition requirement is to compare Z_m across the predefined operator family. Posterior-predictive simulations must reproduce the residual structure without creating false positives in controls, and the conclusion must be stable under reasonable priors on ϵ_m and nuisance widths.

The present paper does not compute these real-data evidences because the required image/model products are not available in the public-only route. It does, however, define the likelihood objects that a future evidence-grade test must instantiate.

16. What Would Count as a Future Proof?

We use the word proof operationally, not metaphysically. A gravitational-lensing proof of the observer principle would require all of the following:

1. A public or auditable image/model product chain for one or more time-delay lenses.
2. A no-T2 model that reproduces a published Ddt or null posterior within a preregistered tolerance.
3. Image labels, positions, parity or parity-computable models, relative delays, and Fermat-potential differences.
4. A T2 perturbation with one shared `epsilon_tau`, not per-object amplitudes.
5. A likelihood-level posterior for `epsilon_tau`, with nuisance parameters marginalized.
6. Evidence or predictive improvement over the no-tau baseline and over nearby operator controls.
7. Prior-sensitivity and posterior-predictive checks.
8. A null result when the T2 channel is removed or scrambled.

The proof condition can be summarized as:

In words, a future proof would require a successful standard no-T2 reproduction, a statistically identified shared `epsilon_tau`, and survival under nuisance, scrambling, and false-positive controls.

This paper satisfies the last three items only in synthetic/control form. It does not satisfy the first two real-data requirements.

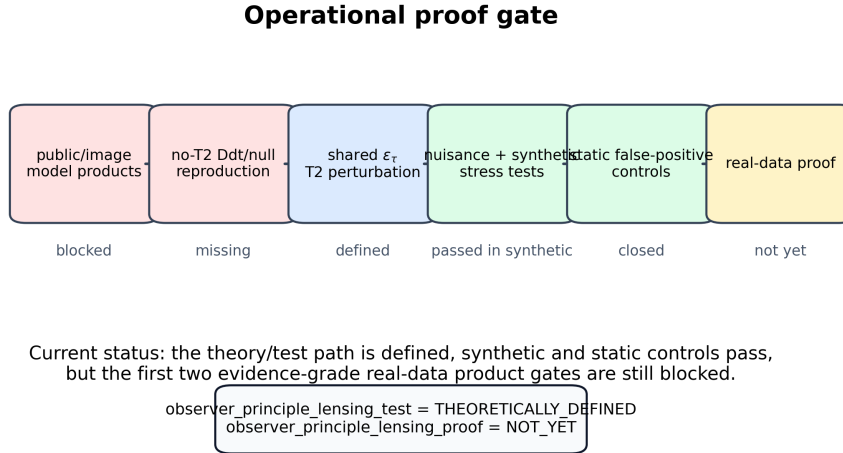


Figure 3: Figure 3. Operational proof gate. The T2 perturbation is not allowed until an evidence-grade no-T2 Ddt/null reproduction is available. Current work defines the test and validates synthetic/static controls, while the first two real-data product gates remain blocked.

17. Why Distance Posteriors Alone Are Insufficient

Published time-delay distance products are valuable cosmography products, but they are too compressed for the observer-gated T2 proof. A Ddt posterior alone does not expose which image parity, Fermat-potential difference, and image-position constraints produced it. Since the T2 channel is image-geometry dependent, the proof must begin upstream of compressed distance chains.

The required product is therefore not merely:

published Ddt posterior

but rather:

image/model products -> no-T2 Ddt reproduction -> controlled T2 perturbation

This distinction is the main practical reason the real-data branch is blocked.

18. Synthetic Identifiability Result

The synthetic branch asks whether the proposed T2 channel is mathematically identifiable before any real-data claim is attempted. The promoted synthetic term was:

T2_parity_abs_fermat

It was promoted only after numerical recovery tests and nearby synthetic controls indicated that it was distinguishable from simpler Fermat-like alternatives.

Nearby controls did not receive the same status:

Term	Role	Status
T1_fermat	ordinary Fermat-like control	expected control fail
T2: parity-weighted absolute Fermat	active tau-core branch	promoted in synthetic tests
T3: path-weighted proxy	proxy branch	not promoted as physics claim

The main hierarchical synthetic result was:

```
lens_count = 28
position_constraint_rows = 224
injected_epsilon_tau = 0.05
epsilon_hat = 0.0499193089
epsilon_error = -0.0000806911
```

Additional synthetic stress results:

```

blind_challenge_scenarios = 4
strict_identified_count = 4
max_abs_epsilon_error = 0.0003890250

```

Noise Monte Carlo:

```

replicates_per_noise = 64
stable_noise_max = 0.008
transition_noise_min = 0.010
fail_noise_min = null

```

Joint stress grid:

```

scenario_count = 8
replicates_per_scenario = 48
stable_count = 6
transition_count = 2
fail_count = 0

```

Interpretation: the synthetic recovery is a generative sanity check, not evidence for new physics. Recovering an injected $\epsilon_\tau = 0.05$ mainly shows that the estimator can recover the signal class it was asked to recover under controlled assumptions. Its value is methodological: it checks identifiability, code behavior, stress degradation, and nearby-control behavior before any real-data claim is attempted. It does not show that nature contains the T2 operator.

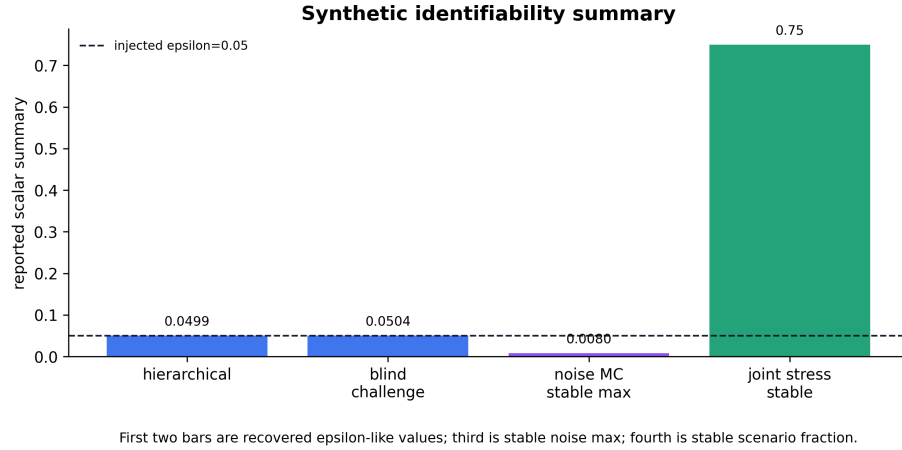


Figure 4: Figure 4. Synthetic identifiability summary. The recovered synthetic epsilon-like values remain near the injected amplitude, while cached stress tests map where stability begins to degrade.

19. Public Time-Delay Product Audit

The public audit asked whether the required proof gate can be run without private communication or unpublished products. The answer was no: the reviewed public products did not provide the target-specific posterior/configuration and Fermat/parity releases needed for this test.

The blocker is:

public products are compressed, paper-only, or software-only rather than target-specific posterior/configuration and Fermat/parity releases

Public no-contact product status for the T2 proof gate

HE0435	yes	partial	no	no
RXJ1131	yes	no	no	no
WFI2033	yes	no	no	no
WGD2038	partial	no	no	no
DES J0408	partial	no	no	no
	public products	image/model posterior	Fermat/parity mapping	no-T2 repro

Blocker: public products are useful, but not evidence-grade for this image-level T2 gate.

Figure 5: Figure 5. Public no-contact product status for the T2 proof gate. The reviewed targets have useful public products, but not the target-specific image/model posterior and Fermat/parity mapping needed for the no-T2 reproduction gate.

19.1 HE0435-1223

HE0435-1223 was the most developed public fallback. Public HST imaging can be ingested, and the reproduction stack defined Ddt tolerance, label conventions, pixel masks, null-model requirements, PSF policy, and archive protocol.

The extracted Ddt tolerance was:

q16/q50/q84 = 2539.56384848 / 2707.4290739999997 / 2889.194643
median tolerance = 54.14858148

However, the public-only model-level PSF correction attempt did not reach an evidence-grade reconstruction:

```
selected_hard_bound_solution_count = 6
baseline_grid_ring_median_rms = 14.4715078608
selected_ring_median_rms = 14.4505423335
improvement_vs_baseline = 0.0209655273
max_ring_five_sigma_fraction = 0.2173664667
```

This is not a criticism of the published HE0435 analysis. It says only that our public-only reconstruction is not evidence-grade enough to support a tau-core perturbation.

19.2 Other Targets

RXJ1131-1231 and WFI2033-4723 were found to have useful public material, but the available products were too compressed for the image-level T2 reproduction required here. WGD2038-4008 and DES J0408-5354 were similarly limited by paper-level or non-target-specific products for this particular test.

These targets remain scientifically valuable. They are blocked only for this specific observer-gated proof gate because the required image/model product depth was not confirmed.

20. Static-Lens Controls

Because real time-delay proof products were unavailable, we developed a static control analysis to test false-positive behavior. The branch used HFF cluster morphology controls: Chandra gas, HFF lensing mass maps, and public member-catalog tracers.

The analysis completed five control layers:

Control layer	Role
HFF morphology selection	select an independent static-lens morphology layer
control scorecard	require heterogeneous controls
null injection	test zero-T2 false positives
null stress sweep	bracket the noise/threshold boundary
systematic-template sweep	flag dangerous templates

The scorecard was heterogeneous:

```
strong_control = 1
mixed_control = 2
weak_negative_control = 1
selected_static_control = macs0717
selected_negative_control = abells1063
```

Baseline null injection:

```
injected_epsilon_tau = 0.0
observed_null_epsilon_hat = 0.004678323836468474
false_positive_rate = 0.0
replicates = 2048
```

Stress boundary:

```
reference_threshold = 0.05
reference_threshold_stable_max_noise = 0.020
first transition at threshold 0.05: noise_sigma = 0.030
```

Template sensitivity:

```
nominal_templates_all_stable = true
inverted_morphology: transition, false_positive_rate = 0.010498046875
residual_aligned_adversarial: fails, false_positive_rate = 0.70947265625
```

The static controls support the falsification discipline of the proof program: the workflow does not automatically see tau-core structure everywhere. But static controls contain no time-delay posterior and therefore cannot prove the observer principle.

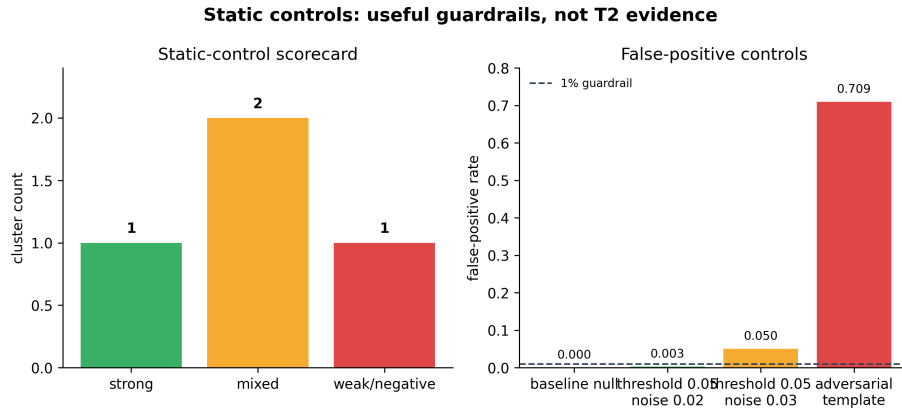


Figure 6: Figure 6. Static-control guardrails. HFF morphology controls remain heterogeneous, pass the baseline null-injection test, and expose noise/template regimes where false positives can be manufactured.

21. Anticipated Reviewer Objections

The most likely reviewer objections are valid, and the paper should be read with them in view.

Objection	Correct response in this paper
The T2 operator looks handcrafted.	Partly. The manuscript now gives a conditional weak-field route, but T2 is still not a derived necessity until the assumptions are obtained from an action, reciprocity theorem, or covariant observer construction.
The synthetic recovery is too clean.	Yes. It is a sanity check showing recovery from a known generative class, not evidence that the class is realized in nature.
There is no real-data posterior.	Correct. No no-T2 Ddt/null reproduction, real-data Bayes factor, or posterior-predictive check is claimed.
The tau-core background theory is too open.	Correct. The paper does not yet specify what fundamental object transforms, what is invariant, or what the primitive measurable is in a complete theory.
The axioms are imposed, not derived.	Correct. The paper now maps their possible origins, but path-reversal evenness, Morse-orientation coupling, and covariant closure remain open theory tasks.
The reciprocity principle is missing.	Partly. The thin-lens reciprocity lemma is proven, but no full covariant reciprocity theorem is claimed.
There is no covariant measurable.	Partly. The paper now proposes an optical measurable candidate using observer proper-time depth and Jacobi-map orientation, but it is not yet derived from a full tau-core action.
The shared amplitude is imposed.	Partly. The paper now gives a tau-accessibility current candidate, but no conservation theorem has yet been proven.

Objection	Correct response in this paper
The current has no action origin.	Partly. The paper gives a minimal Noether-current route from tau shift symmetry, but no complete covariant action is claimed.
The screen construction may be gauge-dependent.	Partly. The paper states screen-gauge covariance conditions, but a full action-level derivation is still missing.
There is no dynamical closure.	Correct. The paper now states the required stress-energy, conservation, and GR-limit conditions, but does not derive them from a complete action.

These objections do not invalidate the narrower contribution. They define its scope. The present manuscript is best classified as a methods and feasibility note with a physics kernel, not as a new-physics detection paper.

22. Main Proposition of the Paper

We can now state the paper’s central theorem in operational form.

Proposition. An operational gravitational-lensing test of the tau-core observer principle is theoretically possible because the observer principle can be mapped to an image-level time-delay lensing operator family with a shared population-level amplitude. The present candidate operator is `T2_parity_abs_fermat`. This proposition does not assert that a complete tau-core action, field content, covariance structure, conservation law, GR-limit derivation, gauge interpretation, or operator-selection principle has been supplied. A future empirical proof would require the T2 amplitude to be identified only after a no-T2 Ddt/null reproduction, likelihood-level nuisance marginalization, evidence comparison against nearby operators, and synthetic and static false-positive controls.

Status. The present work verifies the synthetic and control components of this theorem, but not the real-data no-T2 reproduction component.

Therefore:

```
tau-lensing perturbation test: defined at phenomenological Level 1
real-data proof: not yet achieved
complete tau-core field theory: not yet supplied
dynamical operator-selection principle: not yet supplied
geometrical necessity theorem: not yet supplied
missing ingredient: evidence-grade time-delay image/model products
```

This is the strongest defensible conclusion.

23. Data Used and Reproducibility

The present work used three categories of material.

First, public time-delay lensing product information was audited through local source manifests and product probes. This included HE0435-1223, RXJ1131-1231, WFI2033-4723, WGD2038-4008, DES J0408-5354, H0LiCOW public distance products, TDCOSMO public notebook routes, and related public repositories or metadata pages. These products were sufficient to define blockers but not sufficient to authorize the no-T2 reproduction gate.

Second, synthetic lensing data were used to test whether the proposed T2 channel is identifiable in principle. These tests do not use real lensing measurements as evidence; they test the mathematical behavior of the proposed estimator under controlled noise, nuisance, and position-constraint assumptions.

Third, static HFF cluster products were used as false-positive controls. These include already-generated summaries from Chandra gas, HFF lensing mass maps, and member-catalog morphology controls. They test whether the workflow becomes over-eager in systems without time-delay evidence.

Primary packet:

Primary packet:

```
studies/tau_core_lensing_observer_gate_v01/paper_packet_v01_public_blocked/
```

Critical scripts:

```
repro/scripts/check_repro.py
repro/scripts/build_pdfs.py
studies/tau_core_lensing_observer_gate_v01/make_manuscript_pdf.py
studies/tau_core_lensing_observer_gate_v01/make_arxiv_source.py
```

Critical outputs:

```
outputs/tau_core_lensing_observer_gate_v01/
repro/results/*/summary.json
```

Run:

```
python repro/scripts/check_repro.py
python studies/tau_core_lensing_observer_gate_v01/make_manuscript_pdf.py
python studies/tau_core_lensing_observer_gate_v01/make_arxiv_source.py
python -m pytest -q
```

Expected:

```
REPRO_CHECK_PASS
4 passed
```


24. Limitations

1. No real-data T2 posterior was sampled.
2. No public-only target passed the no-T2 Ddt/null reproduction gate.
3. The HE0435 public fallback failed PSF/model validation and is not a published-posterior reproduction.
4. Synthetic tests validate method behavior, not astrophysical truth.
5. Static controls are false-positive controls, not time-delay evidence.
6. The contact route for original products was disabled by user choice in this work session.
7. The tau-core observer principle is not yet a closed physical theory: action, field content, covariance structure, symmetry, conservation relation, first-principles GR limit, and gauge meaning remain open.
8. The T2 operator is not yet dynamically derived. The paper gives a conditional weak-field route, but its assumptions remain to be derived from a deeper theory and it must be tested against a predefined operator family.
9. No real-data likelihood, evidence, Bayes factor, prior-sensitivity analysis, or posterior-predictive check has yet been computed.
10. No geometrical necessity theorem, reciprocity theorem, variational derivation, or symmetry selection rule has yet been supplied for T2.
11. The paper has not yet defined the primitive physical measurable of $\mathbf{tau}(\mathbf{x}; \mathbf{u})$ in a complete theory: what transforms, what remains invariant, and what detector-accessible quantity carries the tau-core correction.
12. The theorem attempt depends on axioms whose origins are only partially identified; path-reversal evenness, Morse-orientation coupling, conserved tau current, and covariant closure remain open.
13. The reciprocity principle is proven only as a thin-lens lemma. A covariant theorem deriving the measurable, the reversal map, and the current structure has not yet been supplied.
14. No covariant detector-accessible tau-core measurable has yet been derived; the paper lists candidates but does not prove their invariance or thin-lens reduction.
15. The optical measurable $\mathcal{O}_{\tau,i}^{\text{cov}} = \Sigma_i \mathcal{P}(T_i^{\text{cov}})$ is a candidate reduction, not yet a consequence of a full action or conserved tau current.
16. The tau-accessibility current J_{τ}^{μ} is only a closure candidate; no conservation theorem has yet been derived that forces a shared ϵ_{τ} .
17. The Noether-current route assumes a tau shift symmetry and a screen interaction term; neither has yet been derived from a complete covariant tau-core action.
18. Screen-gauge covariance is specified only as a consistency condition; the distinction between physical orientation reversal and arbitrary screen re-labeling remains to be derived from a complete observer construction.
19. Dynamical closure is not derived: the stress-energy contribution, conservation relation, screen response, and GR limit remain requirements for a future complete theory.

25. Conclusion

Gravitational lensing provides a natural route toward testing the tau-core observer principle because a single lens supplies multiple observer-accessible image channels. This paper formulates the required proof criterion and identifies `T2_parity_abs_fermat` as the operative synthetic channel. The channel is identifiable in controlled synthetic tests and is protected by static false-positive controls.

The proof is not yet complete, and the underlying tau-core theory is not yet complete either. The physics kernel is embryonic: no geometrical necessity theorem, symmetry selection rule, reciprocity argument, or variational derivation currently forces T2. Public no-contact products also do not provide the evidence-grade image/model chain needed to reproduce a no-T2 Ddt/null posterior. The line is therefore scientifically alive at the phenomenological test level, but the real-data claim, the operator-necessity claim, and the full physical theory claim remain open.

The next decisive step is not another static control. It is an evidence-grade time-delay lens product chain that permits:

`no-T2 Ddt/null reproduction -> shared-epsilon T2 perturbation -> nuisance and false-positive controls`

Only that sequence can turn the proof program into an empirical proof.

References and Source Anchors

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